CHiMaD

AGENDA

CHiMaD Phase Field Methods XIX

November 4-6, 2025

Scope: This workshop will highlight recent advances in phase field (PF) modeling with a focus on mechanics-driven phenomena. Speakers will explore topics ranging from anisotropic growth, grain evolution to fracture PF, emphasizing both fundamental theory and applications. The program aims to connect researchers both developing and benchmarking new PF methods for complex materials problems.

All times in CST

Tuesday, Nov. 4

12:00PM – 12:40PM **Lunch**

12:40PM – 12:50PM **Welcome & Logistics**
Duncan Burns, *Northwestern University*
Alex Chadwick, *Naval Research Laboratory (NRL)*

12:50PM –01:00PM **Opening Remarks**
Peter Voorhees, *Northwestern University / Caltech*

01:00PM – 02:00PM **Atomistic and Mesoscale Modeling of Grain Boundary Dynamics**
Fadi Abdeljawad, *Northwestern University*

02:00PM – 03:00PM **Regularization of highly anisotropic corners in phase-field theory**
Thomas Philippe, *Ecole Polytechnique*

03:00PM – 03:30PM **Group Photo** (In-person + Online) **& Break**

03:30PM – 03:45PM **LT I: Simulating Twin Evolution in Mg Alloys with the PRISMS-MP Framework**
David Montiel, *University of Michigan*

03:45PM – 04:00PM **LT II: Exploiting Finite Interface Width Effects**
Duncan Burns, *Northwestern University*

04:00PM – 04:15PM **LT III: Phase field simulations of static recovery and recrystallization in Mg-3Zn-0.1Ca with stored energy from PRISMS-plasticity**
Philip Staublin, *University of Michigan*

04:15PM – 04:30PM **Break**

04:30PM – 05:00PM **Grain Growth Benchmark**
Mike Tonks, *University of Florida*

05:00PM – 05:30PM **Open-Ended Discussion**

06:30PM – END **Dinner (at restaurant TBD)**

Talk Formats:
Invited talks – 40-minute presentation + 15-minute questions + 5-minute break
Lightning talks – 10-minute presentation + 5-minute questions

All times in CST

Wednesday, Nov. 5

08:30AM – 08:55AM **Breakfast**

08:55AM – 09:00AM **Opening Remarks**

09:00AM – 10:00AM **Phase-field fracture 2.0: "New formulation!", "Extra strength!"**
Blaise Bourdin, *McMaster University*

10:00AM – 11:00AM **Phase field models to unravel material-environment interactions: hydrogen embrittlement and corrosion**
Emilio Martínez-Pañeda, *University of Oxford*

11:00AM – 11:15AM **Break**

11:15AM – 11:30AM **LT IV: evoxels: A differentiable physics framework for voxel-based microstructure simulations**
Simon Daubner**,** *Imperial College London*

11:30AM – 11:45AM **LT V: Phase Field Modeling of Dead Lithium Formation**
Jin Zhang**,** *Northwestern University*

11:45AM – 12:30PM **Phase Field – TMS Course**
Jin Zhang, *Northwestern University*
David Montiel, *University of Michigan*

12:30PM – 01:30PM **Lunch**

01:30PM – 02:30PM **Coupled phase field frameworks for micromechanical fracture crack propagation in metallic alloys and composites**
Prajwal Arunachala, *Johns Hopkins University*

02:30PM – 03:30PM **Phase-field fracture in fiber reinforced composite structures: anisotropy, mode mixity, delamination in quasi-static and dynamic loading**
Trisha Sain, *Michigan Tech*

03:30PM – 03:45PM **Break**

03:45PM – 04:30PM **Future of PFHub**
Beck Andrews, *University of Michigan*
Philip Staublin, *University of Michigan*
Mythreyi Ramesh, *Northwestern University*

04:30PM – 5:30PM **Lightning Talks: TBD**

05:30PM – END **Dinner**

Talk Formats:
Invited talks – 40-minute presentation + 15-minute questions + 5-minute break
Lightning talks – 10-minute presentation + 5-minute questions

All times in CST

Thursday, Nov. 6

08:30AM – 08:50AM **Breakfast**

08:50AM – 09:00AM **Legacy of CHiMaD**
Jim Warren, *NIST*

09:00AM – 10:00AM **Modeling complex anisotropy in interface energy using the phase field method**
Jacob Bair, *Oklahoma State University*

10:00AM – 10:15AM **Break**

10:15AM – 10:45AM **Direct Willmore regularization of strongly anisotropic interfacial energy**
Beck Andrews, *University of Michigan*

10:45AM – 11:45AM **Recommended Practice Guide (Benchmarks)**

11:45AM – 12:00PM **Break**

11:30AM – 12:30PM **Next Steps: Future of CHiMaD**

01:00PM – END **Lunch**

Talk Formats:
Invited talks – 40-minute presentation + 15-minute questions + 5-minute break
Lightning talks – 10-minute presentation + 5-minute questions

Invited Speaker Biographies:



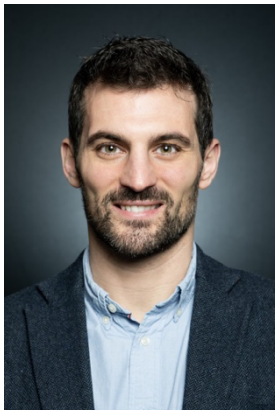
Dr. Fadi Abdeljawad is an Associate Professor in the Department of Materials Science and Engineering at Northwestern University (starting January 2026). Prior to joining Northwestern, he was an Associate Professor at Lehigh University. He obtained his M.A. (2010) and Ph.D. (2014) from Princeton University with a primary focus on theoretical and computational materials science. From 2014-2018, Dr. Abdeljawad was a Staff Scientist in the Computational Materials and Data Science Department at Sandia National Laboratories. Dr. Abdeljawad is a recipient of the 2022 TMS Early Career Faculty Fellow Award; 2020 Army Research Office Young Investigator Program Award; and 2019 Ralph E. Powe ORAU DOE Award. Abdeljawad's research, which is supported by DOD, DOE, NSF, and U.S. National Laboratories, is focused on problems related to microstructure formation and evolution, interfacial dynamics, and mechanics.



Dr Thomas Philippe earned his PhD in physics from Normandy University (Rouen, France), on non-classical nucleation phenomena. He then joined the Voorhees group in 2012 (Northwestern University) as a Postdoctoral Research Fellow to work on coarsening theory and crystallization of amorphous thin films. Since 2015, he is a CNRS Research Fellow in Ecole Polytechnique (Paris, France). His main research interests include nucleation with precursors and regularized phase-field models for faceted crystal growth.



Blaise Bourdin is a Professor of Mathematics at McMaster University (Hamilton, ON, Canada) and the Canada Research Chair in Mathematical and Computational Aspects of Solid Mechanics (tier 1). Prior to moving to McMaster, Bourdin was the A.K. & Shirley Professor of Mathematics at Louisiana State University (Baton Rouge, LA, USA).
Bourdin's formal training is at the confluence of solid mechanics, scientific computing, and applied mathematics. His main interest is in multi-disciplinary modeling, analysis, and numerical simulation in mechanical science with a specific focus on defect mechanics and optimal design. He pioneered the phase-field approach to brittle fracture, one today's most influential method in the field.



Prof Emilio Martinez-Paneda is an Associate Professor at the University of Oxford. Prior to joining Oxford, he was a Reader (Associate Professor) at Imperial College London, where he led an interdisciplinary research group from 2019 to 2023 (2019: Lecturer, 2021: Senior Lecturer, 2023: Reader). Before that, he was an 1851 Research Fellow at the University of Cambridge. His recent research focus has been on addressing fundamental scientific challenges that can accelerate our transition to a net-zero carbon society and assist in tackling our energy and climate crises. Prof Emilio Martinez-Paneda has been the PI on over 7M GBP of funding in the past five years (ERC Starting Grant, UKRI Future Leaders Fellowship) and his work has been recognized through multiple awards, including the 2021 UK Young Engineer of the Year (Royal Academy of Engineering), the 2022 Imperial College President's Medal for Excellence in Research, and the 2024 IACM John Argyris Award.



Dr. Prajwal Kammardi Arunachala is a postdoctoral researcher in the Computational Mechanics Research Laboratory led by Prof. Somnath Ghosh at Johns Hopkins University, having completed his Ph.D. at Stanford University. His current work focuses on micromechanics-informed fatigue modeling in metallic alloys, with an emphasis on high fidelity and performance.
He has received numerous awards, such as the Juan C. Simo Best Thesis Award from Stanford and the Institute Silver Medal, Vidyasagar Nehra Gold Medal, and Prof. Madhav Kulkarni Gold Medal from the Indian Institute of Technology Bombay. His fellowships include the Leavell and Stanford School of Engineering Graduate Fellowships from Stanford, the S.C.Mehrotra and Institute Academic Prizes from IIT Bombay, and the National Talent Search and KVPY scholarships from the Government of India.



Dr. Trisha Sain is currently an Associate Professor in the Mechanical and Aerospace Engineering department at Michigan Tech. She joined in August 2016 as an Assistant Professor. Prior to joining Michigan Tech, she was an Assistant Professor in the Department of Mechanical Engineering, North Carolina A&T State University, Sep 2013 - Aug 2016. She has Post-doctoral experience at University of Michigan, Ann Arbor, 2011-2013, Mechanical Engineering and Aerospace Engineering, and Post-doctoral experience at Technical University of Catalunya, Barcelona, Spain, 2009-2011. She holds a PhD from Indian Institute of Science Bangalore (2008). Her research focuses on understanding the behavior of advanced engineering materials based on the principle of solid mechanics. The current emphasis is on-

- multiscale and coupled physics behavior of polymers and polymer composite materials.
- fracture, damage and impact behavior of materials.
- degradation of metallic implants in biomedical applications
- uncertainty quantification of physics-based modeling.

Jacob L. Bair was raised on a farm near Ephrata, WA. He earned a B.S. degree in Mechanical Engineering with a second major in French from Washington State University in December of 2012, then his Ph.D. in Materials Science and Engineering from Missouri University of Science and Technology in December of 2016. His research has been on experimental and computational studies of the microstructural evolution of materials using scanning electron microscopy with Electron Backscatter Diffraction, Phase Field Modeling, and Molecular Dynamics. Topics he has researched include hydrides in Zr fuel rod claddings, grain boundary migration in face centered cubic materials, and twinning in nickel. He spent several years at Pacific Northwest National Laboratory before joining Oklahoma State University in 2020. He maintains a joint appointment at PNNL and is currently an Assistant Professor at Oklahoma State University in the Mechanical and Aerospace Engineering Department.

